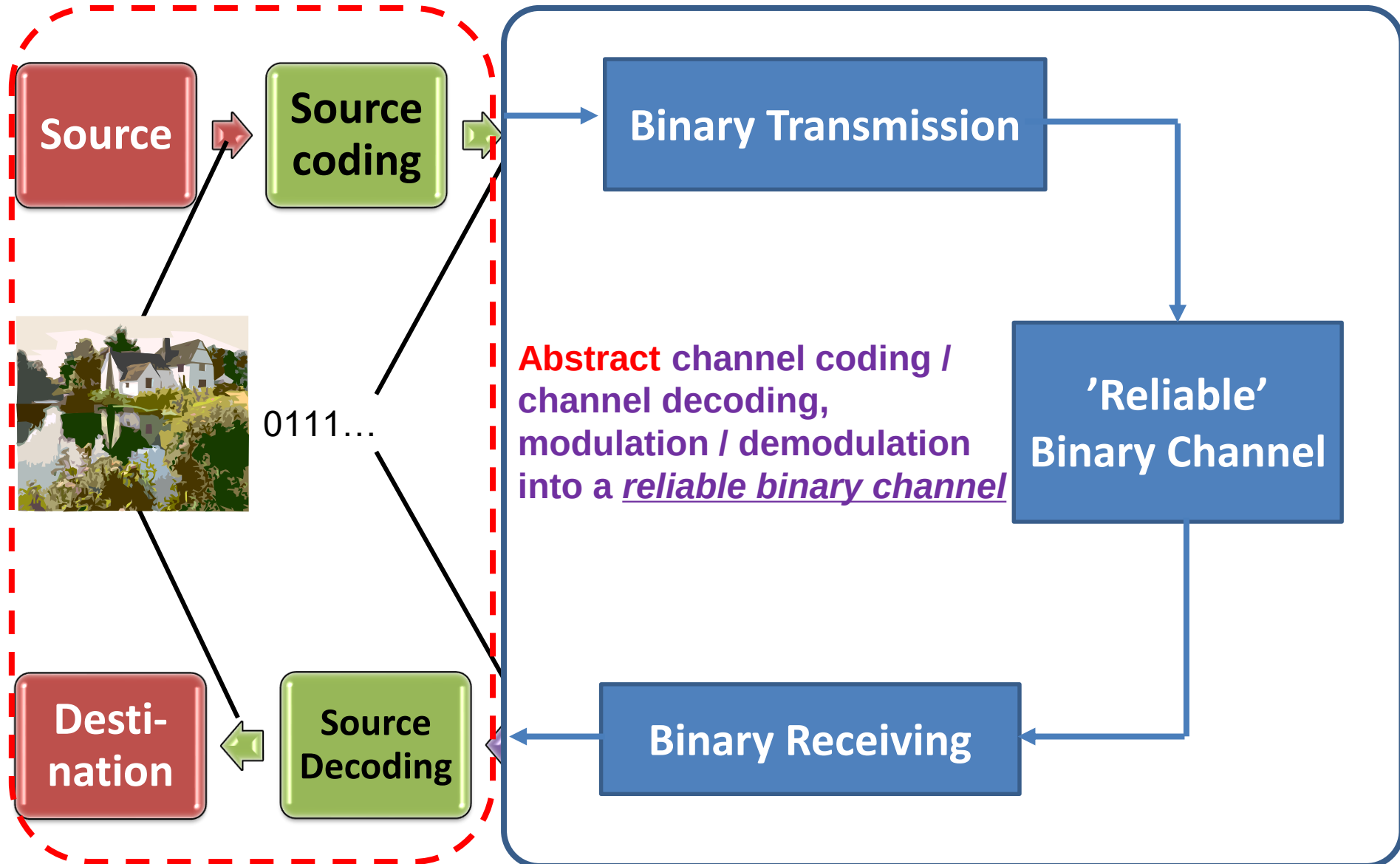


Communications and Networks

Sampling and Quantization

Sheng Zhou

Design Perspective of Source Coding





Source Coding: Overview

This Chapter: Digitalization of Analog Source

Analog Input

Sampling

Quantization

Sampled Signal
(Discrete-Time,
Continuous-Value)

Analog Output

Analog Filter

Reconstruction

Last Chapter: Information Theory and Source Coding

Compression Encoder

Bit sequence
(0, 1)

'Reliable'
Binary Channel

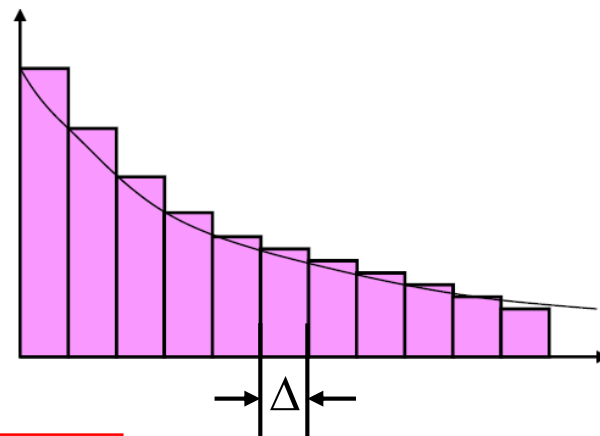
Compression Decoder

Bit sequence
(0, 1)

Previously...

- For continuous-valued source,
- Differential entropy

$$H(X) = -\sum_i p_i \log p_i$$



$$\begin{aligned} H(X) &= E \left\{ H(x < X \leq x + \Delta) \right\} = E \left\{ \log \frac{1}{p(x)\Delta} \right\} \\ &= -\int_{-\infty}^{+\infty} p(x) \log p(x) dx + \lim_{\Delta \rightarrow 0} \log \frac{1}{\Delta} \end{aligned}$$

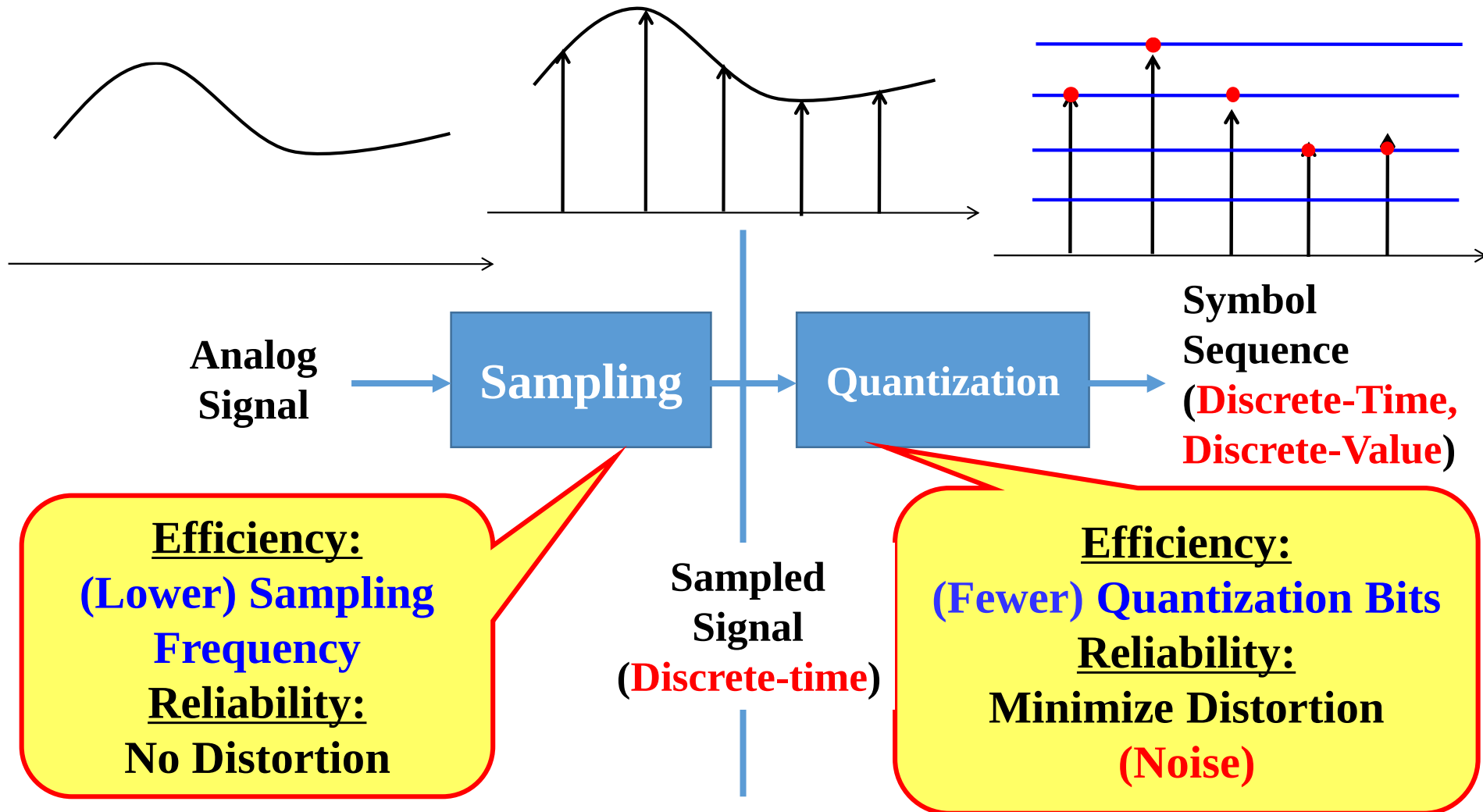
Tends to infinity!

- Further, what if the signal is both continuous in value and time?
- e.g. images, videos, audios, etc.

Introduction (1)

- **Sampling and quantization in digital communications**
 - Some sources are naturally discretized (e.g. characters)
 - Others are analog sources, which needs discretization
 - Time-discretization - Sampling
 - Value-discretization - Quantization

Introduction (2)



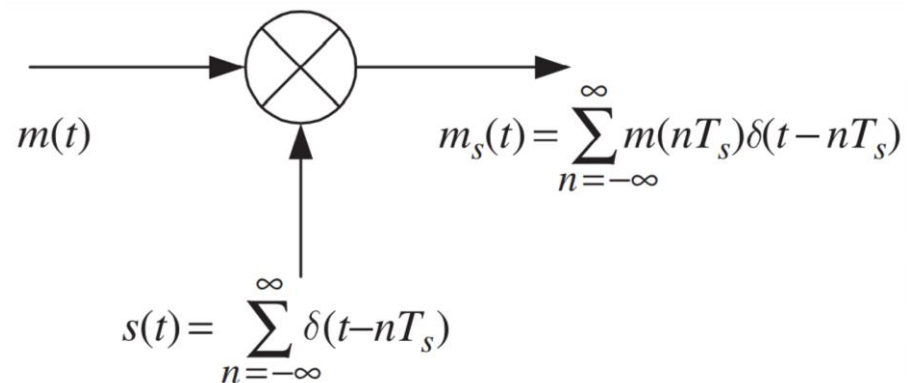
Ideal Sampling (1)

- **Mathematical Description**

- The simplest sampling model
- Analog waveform $m(t)$ is multiplied by a periodic sequence of unit impulse functions $s(t)$

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- Illustration:



- T_s : sampling period
- $f_s = 1/T_s$: sampling frequency

Ideal Sampling (2)

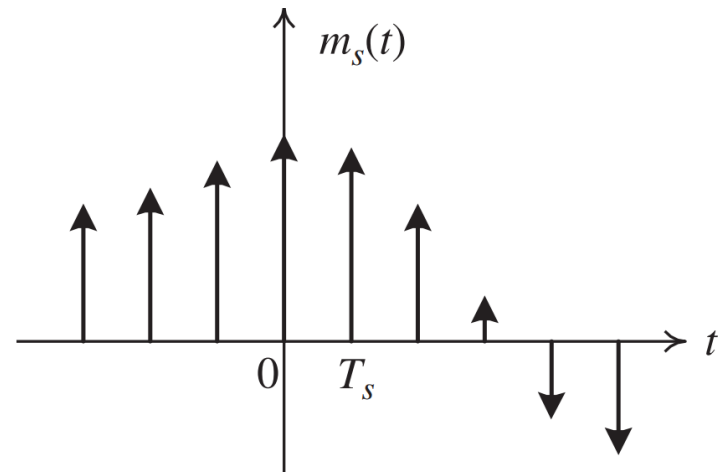
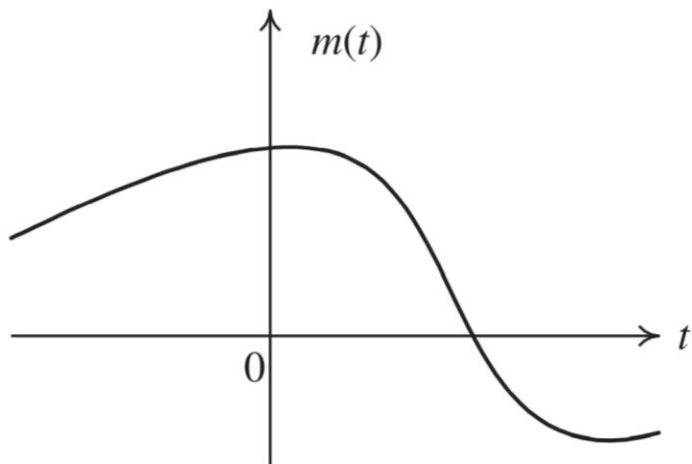
- **Mathematical Description**

- The sampled waveform can be expressed as

$$m_s(t) = m(t)s(t)$$

$$= \sum_{n=-\infty}^{\infty} m(t)\delta(t - nT_s)$$

$$= \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t - nT_s)$$



Ideal Sampling (3)

- **Fourier Analysis**

- The Fourier transform of an impulse sequence $s(t)$ is another impulse sequence $S(f)$

$$S(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

- Thus the transform of the sampled waveform is

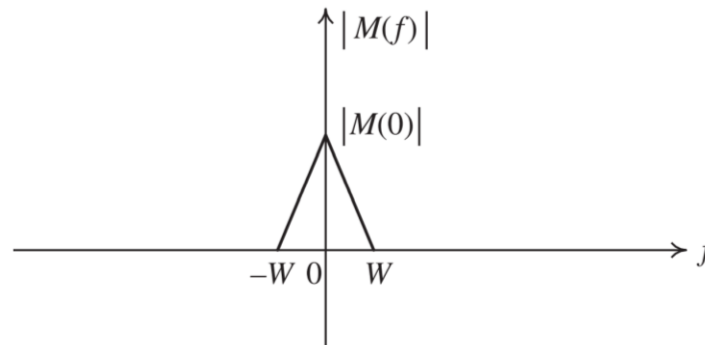
$$\begin{aligned} M_s(f) &= M(f) * S(f) \\ &= \frac{M(f)}{T_s} * \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \\ &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} M(f - nf_s) \end{aligned}$$

- The spectrum of sampled waveform consists of an infinite number of shifted copies of the original spectrum

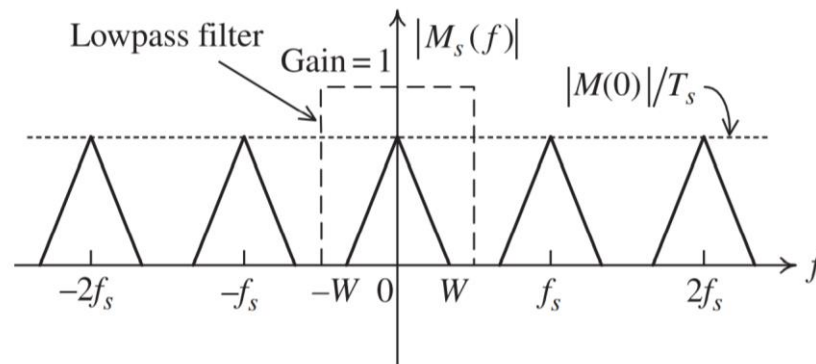
Sampling for Lowpass Signal (1)

- **Reconstruction without Aliasing**

- For lowpass signal, $M(f)$ is zero outside $[-W, W]$



- If $f_s \geq 2W$, there is no aliasing (overlapping)
- Through an idealized lowpass filter, the original signal can be reconstructed



Sampling for Lowpass Signal (2)

- **Sampling Theorem (Lowpass signal)**

- A signal restricted to $[-W, W]$ is completely described by the value of the signal at periodic time instants that are separated by at most $1/(2W)$ seconds
- **Nyquist Rate:** $f_s = 2W$
- **Nyquist Criterion:** $f_s \geq 2W$

- **Waveform Reconstruction**

- The sampled waveform

$$m_s(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t - nT_s)$$

- Apply Fourier transform

$$M_s(f) = \mathcal{F}\{m_s(t)\}$$

Sampling for Lowpass Signal (3)

- **Waveform Reconstruction**

- Rewrite as follows

$$\begin{aligned} M_s(f) &= \sum_{n=-\infty}^{\infty} m(nT_s) \mathcal{F}\{\delta(f - nf_s)\} \\ &= \sum_{n=-\infty}^{\infty} m(nT_s) e^{-j2\pi n f T_s} \end{aligned}$$

- Since $M(f) = M_s(f)/f_s$, $-W \leq f \leq W$, we have

$$M(f) = \frac{1}{f_s} \sum_{n=-\infty}^{\infty} m(nT_s) e^{-j2\pi n f T_s}$$

Sampling for Lowpass Signal (4)

- **Waveform Reconstruction**

- By inverse Fourier transform,

$$\begin{aligned} m(t) &= \mathcal{F}^{-1}\{M(f)\} \\ &= \int_{-\infty}^{\infty} M(f) e^{j2\pi ft} df \\ &= \int_{-W}^W \frac{1}{f_s} \sum_{n=-\infty}^{\infty} m(nT_s) e^{-j2\pi n f T_s} e^{j2\pi ft} df \\ &= \frac{1}{f_s} \sum_{n=-\infty}^{\infty} m(nT_s) \int_{-W}^W e^{j2\pi f(t-nT_s)} df \\ &= \sum_{n=-\infty}^{\infty} m(nT_s) \frac{\sin[2\pi W(t-nT_s)]}{\pi f_s(t-nT_s)} \end{aligned}$$

Sampling for Lowpass Signal (5)

- **Waveform Reconstruction**

- Let $f_s = 2W$ (Nyquist rate), we have

$$\begin{aligned} m(t) &= \sum_{n=-\infty}^{\infty} m\left(\frac{n}{2W}\right) \frac{\sin(2\pi Wt - n\pi)}{2\pi Wt - n\pi} \\ &= \sum_{n=-\infty}^{\infty} m\left(\frac{n}{2W}\right) \text{sinc}(2Wt - n) \end{aligned}$$

- where

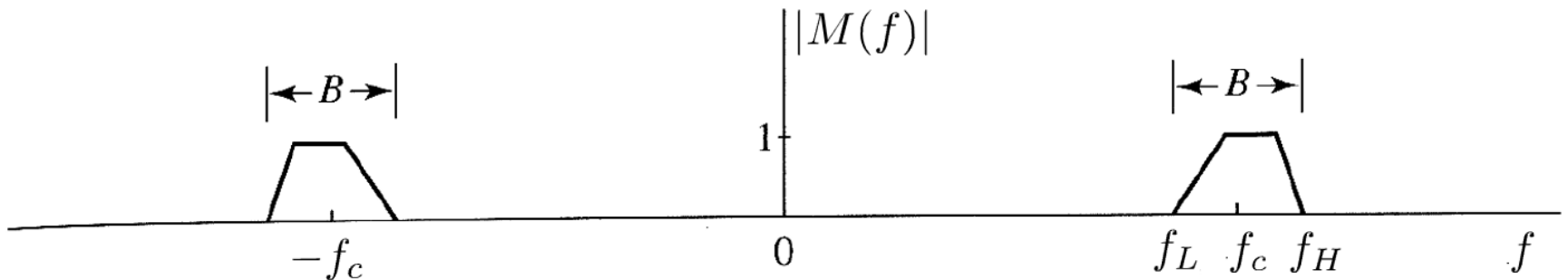
$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

- This provides an **interpolation** formula for the reconstruction of the original signal from its sampled values.

*Sampling for Bandpass Signal (1)

- **Bandpass Signals**

- Only has frequency contents in $[-f_H, -f_L] \cup [f_L, f_H]$
- Bandwidth: $B = f_H - f_L$



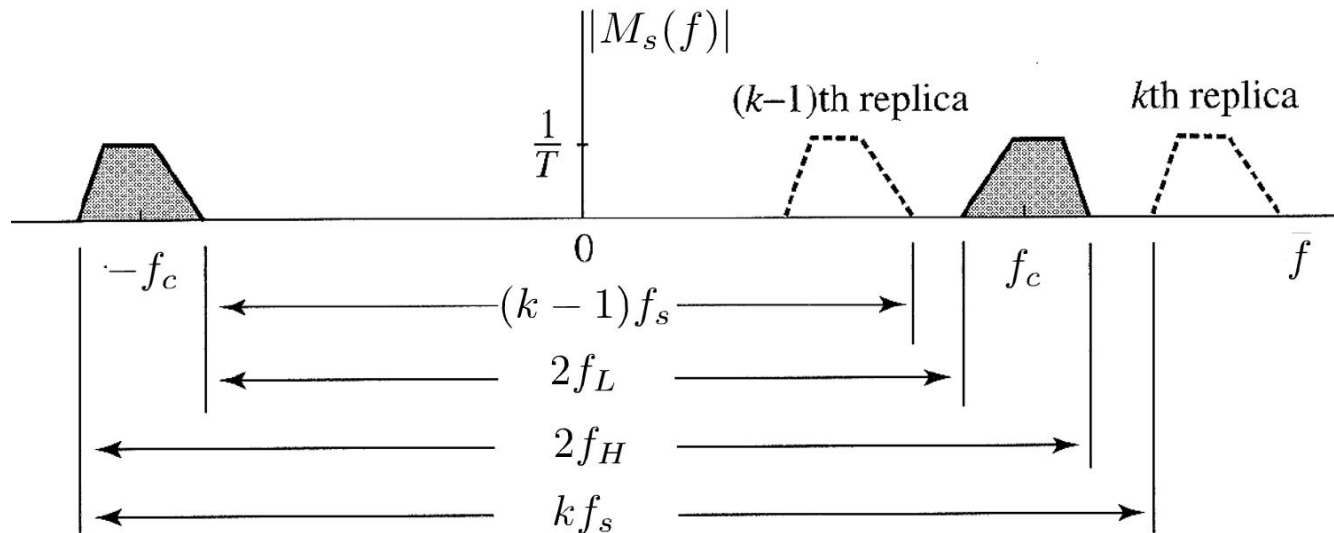
- If we naively apply sampling theorem for lowpass signal, the Nyquist rate is $2f_H$
- However, there are sampling techniques that allow sample rate consistent with bandwidth B , rather than f_H

*Sampling for Bandpass Signal (2)

- **Reconstruction without Aliasing**

- The spectrum of sampled waveform consists of an infinite number of shifted copies of the original spectrum
- Condition: the $(k - 1)$ th and k th shifted copies of the negative spectral band do not overlap with the positive band

$$2f_H \leq kf_s \quad \& \quad (k - 1)f_s \leq 2f_L$$



*Sampling for Bandpass Signal (3)

- **Reconstruction without Aliasing**

- We can see the sampling frequency should be in the range

$$\frac{2f_H}{k} \leq f_s \leq \frac{2f_L}{k-1}$$

- Let $B = f_H - f_L$ and we solve the inequality for k

$$\begin{cases} \frac{1}{f_s} \leq \frac{k}{2f_H} \\ (k-1)f_s \leq 2f_H - 2B \end{cases} \Rightarrow k \leq \frac{f_H}{B}$$

- Therefore, the maximum value of k is obtained

$$1 \leq k \leq k_{max} = \left\lfloor \frac{f_H}{B} \right\rfloor$$

*Sampling for Bandpass Signal (4)

- **Feasible Sampling Rates**

- The feasible sampling rates for reconstruction without aliasing is the union of k_{max} intervals
 - The interval corresponding to $k = 1$ is the sampling theorem for lowpass signals

$$2f_H \leq f_s \leq \infty$$

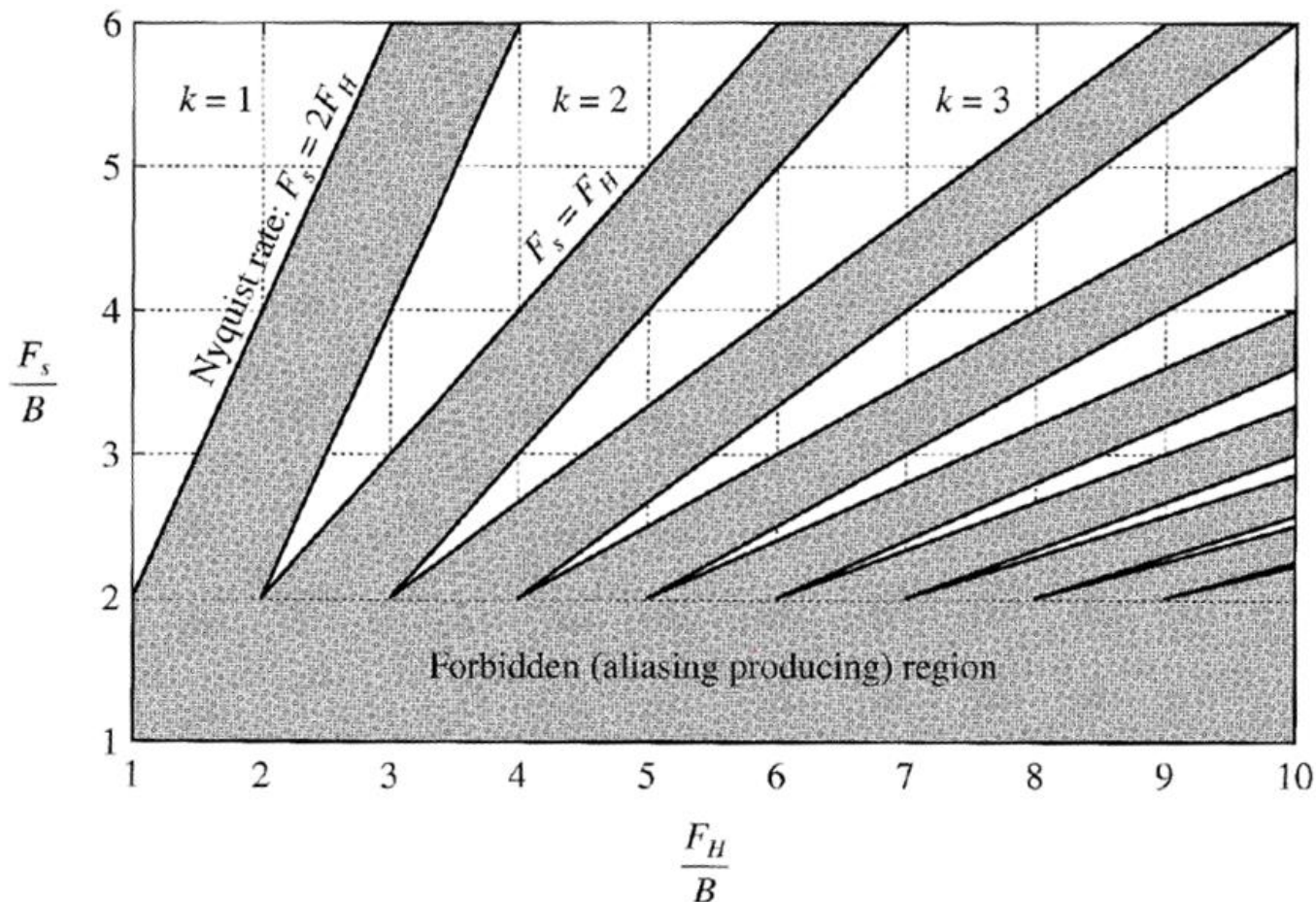
- Normalizing by bandwidth, the feasible sampling rate is a function of the band position f_H/B

$$\frac{2}{k} \frac{f_H}{B} \leq \frac{f_s}{B} \leq \frac{2}{k-1} \left(\frac{f_H}{B} - 1 \right)$$

*Sampling for Bandpass Signal (5)

- **Feasible Regions**

- The lowest points of the wedges corresponds to $f_s = 2B$



*Sampling for Bandpass Signal (6)

- **Sampling Theorem (Bandpass signal)**

- Given the frequency band of the bandpass signal, the minimum sampling frequency is

$$f_{s,min} = \frac{2f_H}{k_{max}} = 2B \left(\frac{f_H}{B} \right) \frac{1}{k_{max}}$$

- Introduce

$$\left\{ \frac{f_H}{B} \right\} = \frac{f_H}{B} - k_{max} = \frac{f_H}{B} - \left\lfloor \frac{f_H}{B} \right\rfloor$$

- Therefore,

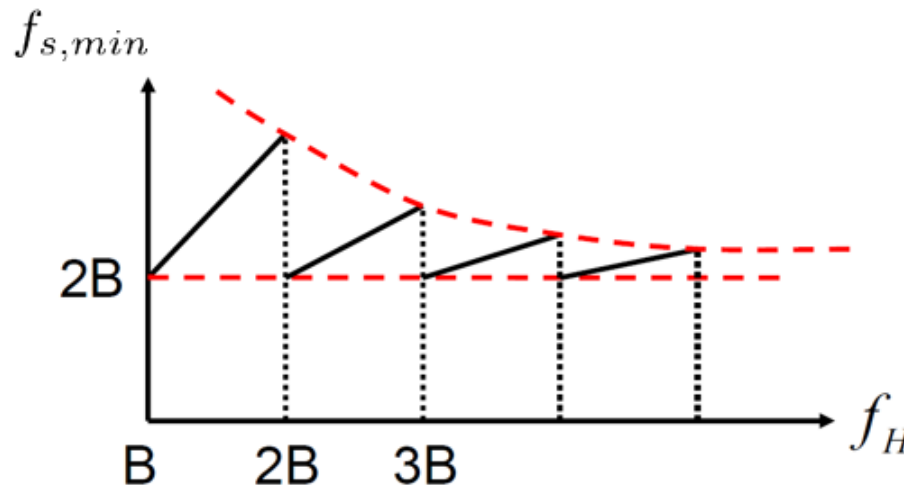
$$f_{s,min} = 2B \left(1 + \frac{\{f_H/B\}}{k_{max}} \right)$$

- where k_{max} and $\{f_H/B\}$ are the integer and decimal parts of f_H/B

Sampling for Bandpass Signal (7)

- **Sampling Theorem (Bandpass signal)**

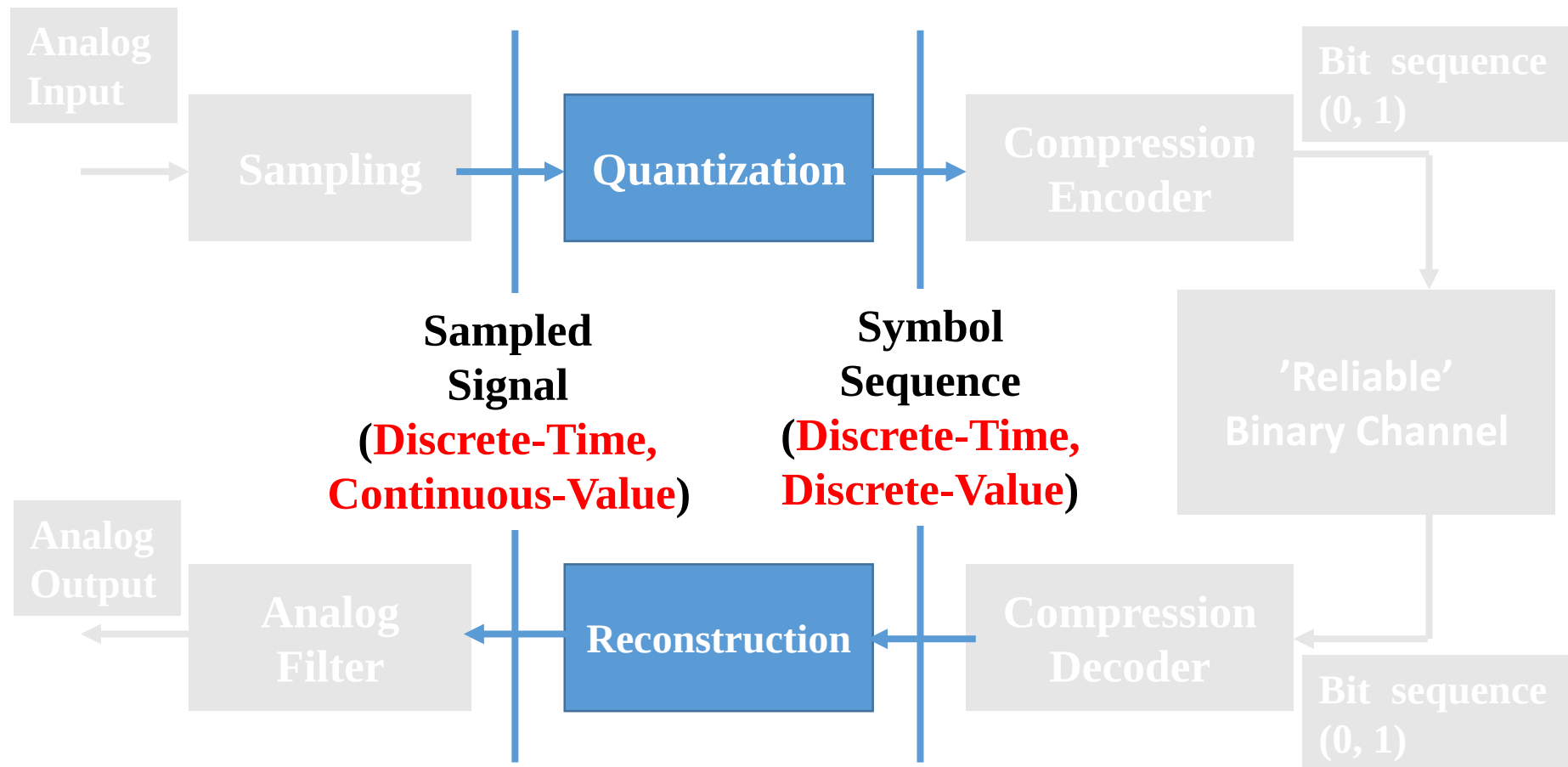
- The minimum sampling frequency is a function of f_H



- The lower envelope $f_{s,min} \geq 2B$ with equality when f_H/B is an integer
- The upper envelope is

$$f_{s,min} \leq 2B \left(1 + \frac{1}{f_H/B - 1} \right) = 2B \left(1 + \frac{B}{f_H - B} \right)$$

Source Coding: Overview



Quantization (1)

- **Quantization**

- To obtain a fully digital representation of a continuous signal, two further operations are needed:
 - Quantization of the amplitude of the sampled signal
 - Encoding of the quantized values

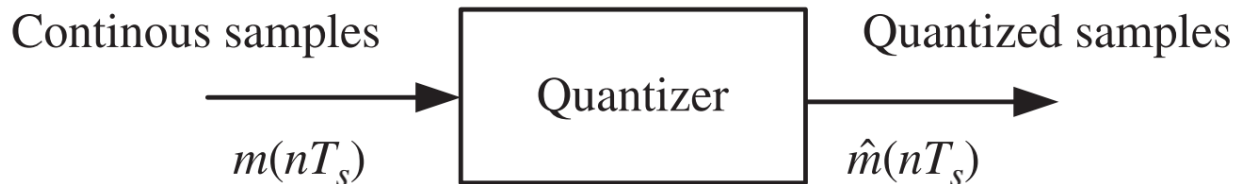


- **Quantization:** the process of transforming the sample amplitude of a message signal into a discrete amplitude taken from a finite set of possible amplitude
- Unlike reconstruction from sampling, there **is always information loss** for quantization process

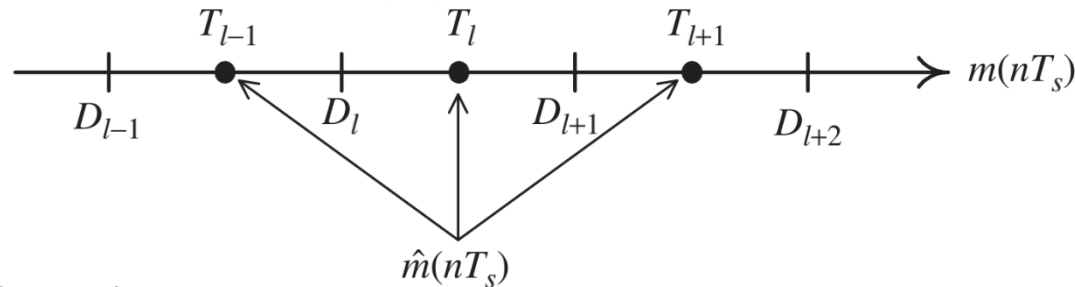
Quantization (2)

- **Basic Model of Quantization**

- Assume the quantization process is memoryless and instantaneous
 - The quantization of sample value at time nT_s is independent of earlier or later samples
- With this assumption, the input and output of quantization can be simplified from $m(nT_s)$, $\hat{m}(nT_s)$ to m , $\hat{m}$



Quantization (3)



- **Basic Model of Quantization**

- Decision Levels:
- Let the amplitude range of the continuous signal be partitioned into L intervals, where the l 's interval is determined by decision levels D_l, D_{l+1}

$$\mathcal{I}(l) : \{D_l < m \leq D_{l+1}\}, \quad l \in \{1, \dots, L\}$$

- Step Sizes: $\Delta_l = D_{l+1} - D_l$
- Representation Point: represents all the signal amplitudes in the interval $\mathcal{I}(l)$ by some amplitude T_l

$$\mathcal{Q}(m) = \mathcal{Q}(\{D_l < m \leq D_{l+1}\}) = T_l$$

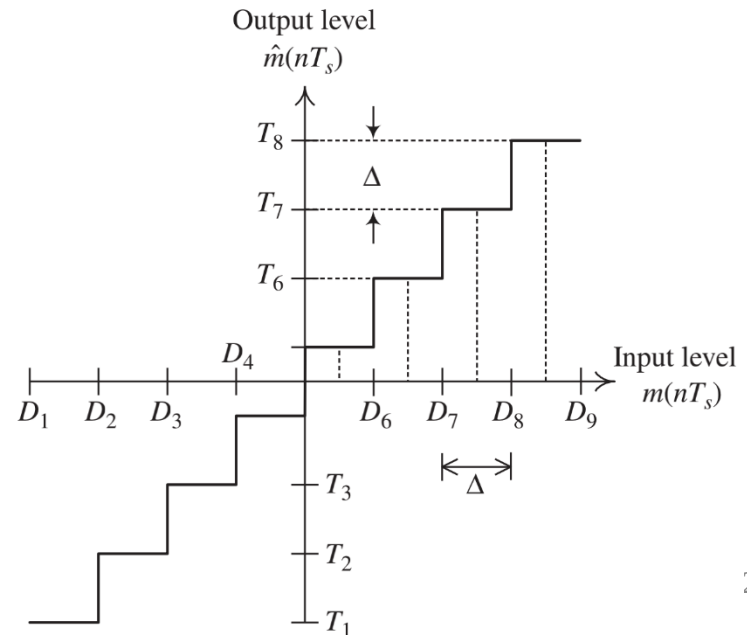
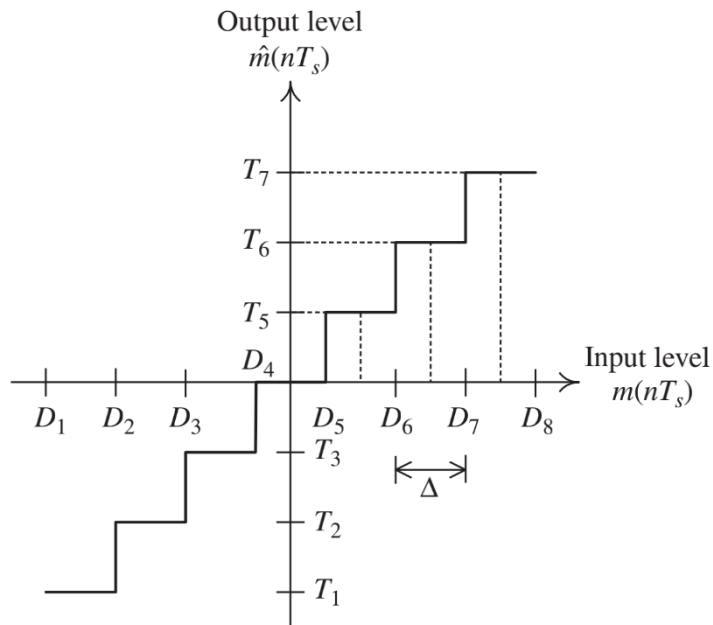
Uniform Quantizer (1)

- **Introduction**

- Definition: Step sizes is the same for each interval

$$\Delta_l \equiv \Delta \quad \forall l \in \{1, \dots, L\}$$

- Midtread: the origin lies in the middle of a tread
- Midrise: the origin lies in rise of the staircase



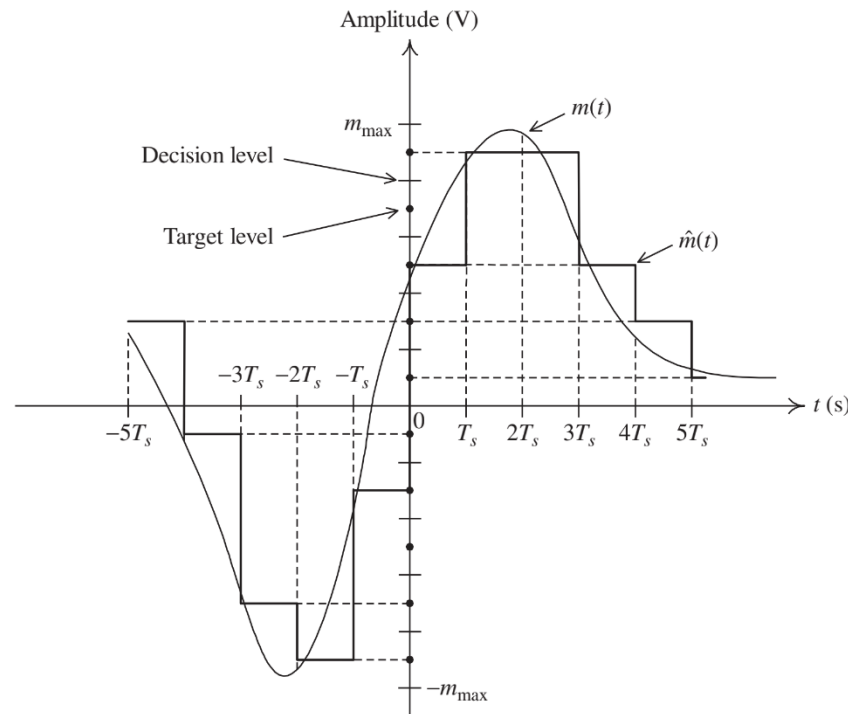
Uniform Quantizer (2)

- **Introduction**

- For both characteristics, the decision levels are equally spaced and the l 's representation point is the midpoint of l 's interval

$$T_l = \frac{D_l + D_{l+1}}{2}$$

- Example: the input and output of a midrise uniform quantizer



Uniform Quantizer (3)

- **SNR Analysis**

- Quantization process always introduces an error
- The performance of a quantizer is evaluated in terms of SNR
- Assume the input is a zero-mean r.v. m with p.d.f. $f_m(m)$ with range $-m_{\{max\}} < m < m_{\{max\}}$, then the step size is

$$\Delta = \frac{2m_{max}}{L}$$

- Let $q = m - \hat{m}$ be the error introduced by quantizer, then
$$-\Delta/2 \leq q \leq \Delta/2$$
- If Δ is sufficiently small, it is reasonable to assume the error is a uniform r.v. on $[-\Delta/2, \Delta/2]$ (why?)

$$f_q(q) = \begin{cases} 1/\Delta & -\Delta/2 \leq q \leq \Delta/2 \\ 0 & \text{otherwise} \end{cases}$$

Uniform Quantizer (4)

- **SNR Analysis**

- The mean of the error is zero, and the variance is

$$\sigma_q^2 = \int_{-\Delta/2}^{\Delta/2} q^2 f_q(q) dq = \int_{-\Delta/2}^{\Delta/2} q^2 \left(\frac{1}{\Delta}\right) dq = \frac{\Delta^2}{12} = \frac{m_{max}^2}{3L^2}$$

- Since the target level is typically encoded in binary form, the number of quantization errors is usually chosen to be $L = 2^R$ for convenience. Substituting L ,

$$\sigma_q^2 = \frac{m_{max}^2}{3 \times 2^{2R}}$$

- The average power of the message is equal to the variance of m : (if not overloaded)

$$\sigma_m^2 = \int_{-m_{max}}^{m_{max}} m^2 f_m(m) dm$$

Uniform Quantizer (5)

- **SNR Analysis**

- Therefore SNR_q is expressed as

$$\text{SNR}_q = \left(\frac{3\sigma_m^2}{m_{max}^2} \right) 2^{2R} = 3 \times 2^{2R} D^2,$$

- where D is the **normalized effective value** of signal $m(t)$:

$$D = \frac{\sigma_m}{m_{max}}$$

- We can conclude that the SNR_q of a uniform quantizer increases exponentially with the number of bits per sample R and increases with the normalized effective value D .
 - D is an inherent property of the signal source while R is a technical specification, can be increased to improve SNR_q

Uniform Quantizer (6)

- **SNR Analysis**

- Expressed in decibels (dB):

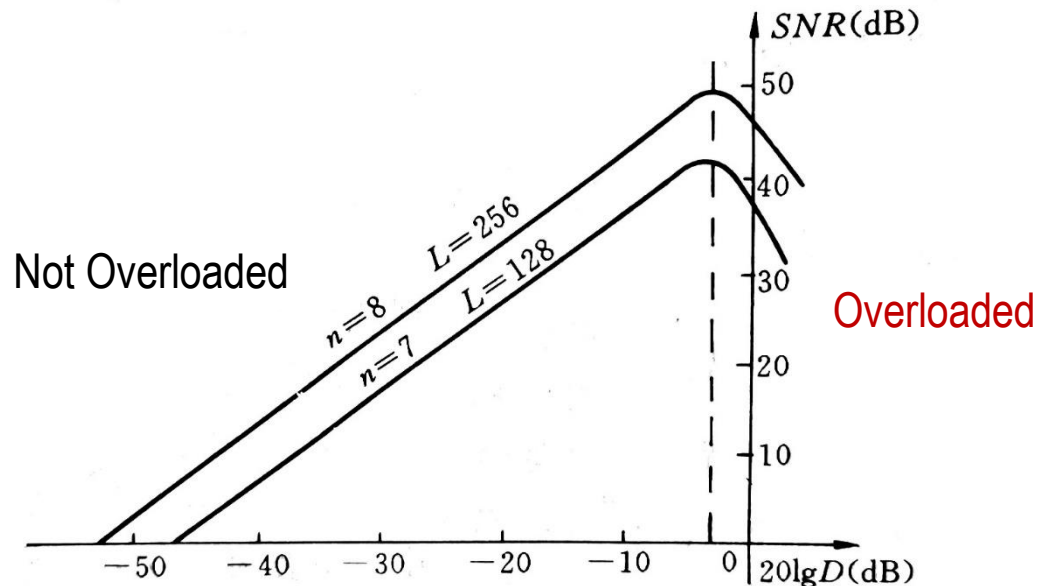
$$\begin{aligned} 10 \log_{10} \text{SNR}_q &= 6.02R + 10 \log_{10} \left(\frac{\sigma_m^2}{m_{max}^2} \right) + 4.77 \\ &= 6.02R + 20 \log_{10} D + 4.77 \end{aligned}$$

- The above equation ‘6dB rule’ points out: an additional 6dB improvement in SNR_q is obtained for each bit added to represent the continuous signal sample

Uniform Quantizer (7)

- **SNR Analysis**

- The total noise σ_{qs}^2 is the sum of normal noise σ_q^2 and overload noise σ_{q0}^2
- Below is the SNR_q characteristics of uniform quantizer. When overloaded, the overload noise increases linearly with the signal power



Uniform Quantizer (8)

- SNR Analysis (**Overload Noise**)

- In the previous analysis, we assume the maximum decision level $V_{max} = m_{\{max\}}$, the minimum decision level $V_{min} = -m_{\{max\}}$
- However in reality, the range of the source may be out of prior estimation, resulting in an overload noise σ_{q0}^2 , which deteriorates the performance of quantizer

$$\sigma_{q0}^2 = \int_{V_{max}}^{\infty} \left[m - \left(V_{max} - \frac{\Delta}{2} \right) \right]^2 f_m(m) dm + \int_{-\infty}^{V_{min}} \left[m - \left(V_{min} + \frac{\Delta}{2} \right) \right]^2 f_m(m) dm$$

Optimal Quantizer (1)

- **Definitions**

- Though simple, uniform quantizers are not optimal in terms of minimizing the SNR_q
- Consider a message signal $m(t)$, let $[-m_{\max}, m_{\max}]$ be the amplitude range of message, partitioned into L quantization regions with $L + 1$ decision levels D_l
- The decision levels are constrained to satisfy:

$$\begin{cases} D_1 = -m_{\max}, \\ D_{L+1} = m_{\max}, \\ D_l \leq D_{l+1} \quad \forall l \in \{1, \dots, L\}. \end{cases}$$

- A representation point T_l may lie anywhere **within** its quantization region

Optimal Quantizer (2)

- **Definitions**

- The average quantization noise power is given by

$$N_q = \sum_{l=1}^L \int_{D_l}^{D_{l+1}} (m - T_l)^2 f_m(m) dm.$$

- Optimal quantization problem: find the set of $2L + 1$ variables $\{D_l, T_l\}$ to maximize the SNR_q , or equivalently minimize the average power of the quantization noise

$$\arg \min_{D_1 \leq T_1 \leq D_2 \leq T_2 \leq \dots D_L \leq T_L \leq D_{L+1}} N_q$$

- An analytical solution to the optimal quantizer design is not possible but by differentiating N_q we could have two basic conditions

*Optimal Quantizer (3)

- **Solving the Optimal Quantization Problem**
 - Leibniz's rule: for integral

$$\int_{a(x)}^{b(x)} f(x, t) dt, \quad -\infty < a(x), b(x) < \infty$$

- The derivative is follows:

$$\begin{aligned} \frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) &= f(x, b(x)) \frac{d}{dx} b(x) - f(x, a(x)) \frac{d}{dx} a(x) \\ &\quad + \int_{a(x)}^{b(x)} \frac{d}{dx} f(x, t) dt \end{aligned}$$

Optimal Quantizer (4)

- **Solving the Optimal Quantization Problem**

- Using Leibniz's rule, differentiating with respect to decision levels and target levels leads to:
- 1. The decision levels are the midpoints of adjacent representation points

$$\frac{\partial N_q}{\partial D_j} = f_m(D_j) [(D_j - T_{j-1})^2 - (D_j - T_j)^2] = 0$$
$$\Rightarrow D_l^{\text{opt}} = (T_{l-1} + T_l)/2$$

- 2. Each representation point is the centroid of its quantization region

$$\frac{\partial N_q}{\partial T_j} = -2 \int_{D_j}^{D_{j+1}} (m - T_j) f_m(m) dm = 0$$
$$\Rightarrow T_l^{\text{opt}} = \frac{\int_{D_l}^{D_{l+1}} m f_m(m) dm}{\int_{D_l}^{D_{l+1}} f_m(m) dm}$$

Optimal Quantizer (5)

- **Solving the Optimal Quantization Problem**

- The derivations give the necessary and sufficient conditions for a quantizer to be optimal, a.k.a. **Lloyd-Max conditions**
- When the source has uniform distribution, the centroid points are midpoints, then uniform quantizer is optimal
- For general source, the optimal quantizer is designed in an iterative manner as follows:

- 1. Start by specifying an arbitrary set of decision levels

$$D_1 \leq \dots \leq D_{L+1}$$

- 2. Find the representation points T_l using condition 2
 - 3. Determine the new decision levels D_l using condition 1
 - 4. Step 2 and 3 are iterated, until the parameters do not change significantly (converged)

Optimal Quantizer (6)

- **Disadvantages of Optimal Quantizer**

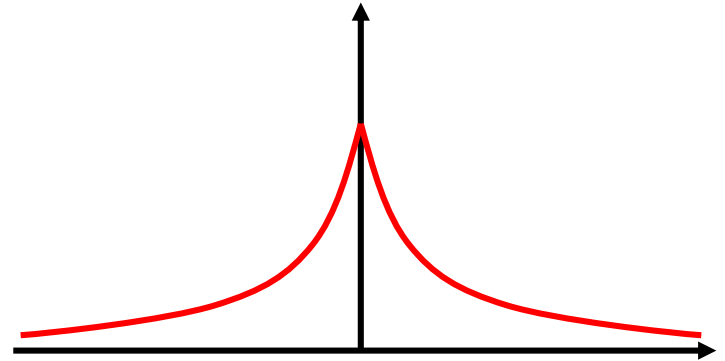
- Though optimal in quantization SNR, the optimal quantizer is not useful in practical applications.
- In practice, the quantizer in use may have to deal with **a variety of sources**, whose statistical properties (p.d.f) are difficult to estimate. In addition, the optimal quantizer is **sensitive to signal levels $m_{\max}$** and has **poor robustness**.

Robust Quantizer (1)

- **Background**

- For certain signals, the input distribution is far from uniform. For a voice signal, there exists a higher probability for smaller amplitudes (such as silent periods and soft speech)

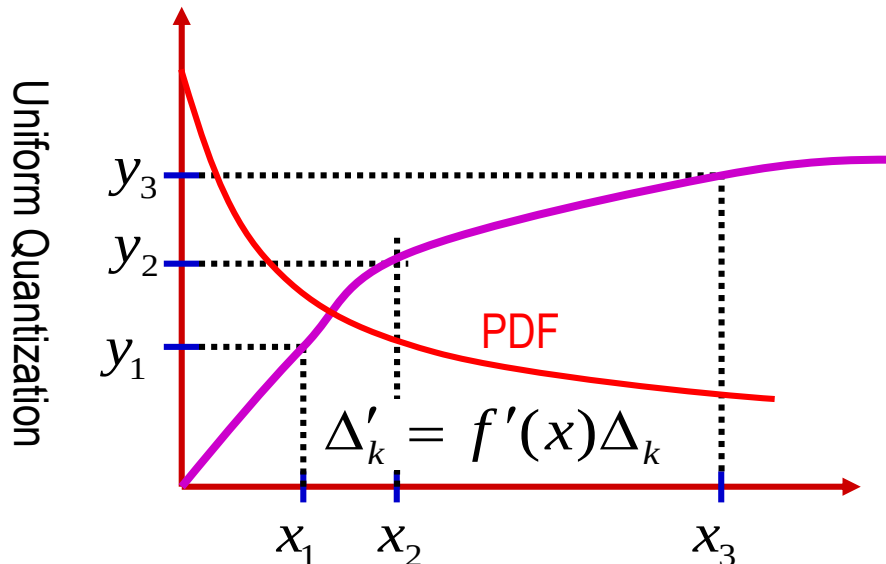
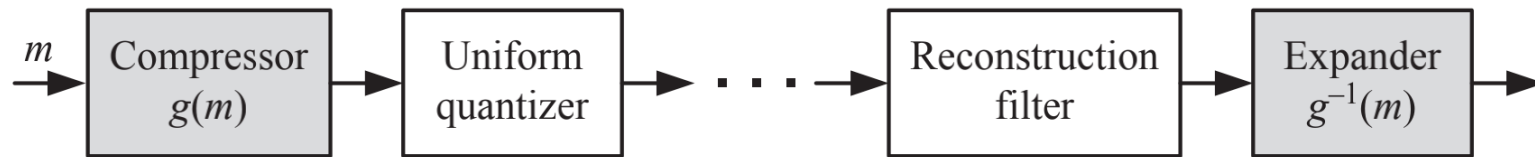
$$p(x) = \frac{1}{\sigma_x \sqrt{2}} e^{-\frac{\sqrt{2}|x|}{\sigma_x}}$$



- For voice signal quantization, it is more efficient to design a quantizer with more quantization regions at lower amplitudes and less quantization regions at larger amplitudes, resulting in a **non-uniform quantizer**

Robust Quantizer (2)

- **Construction of a Non-Uniform Quantizer**
 - Reuse the uniform quantizer!
 - **Compander** = Compressor + Expander



Equivalently we have smaller quantization intervals (higher quantization precision) when the value of the PDF is higher (higher PDF is).

Quite intuitive!

Robust Quantizer (3)

- **A-law and μ -law**

- There are **two types of compander** (compressor + expander) that are widely used for voice signals
- The compression characteristic of A-law is:

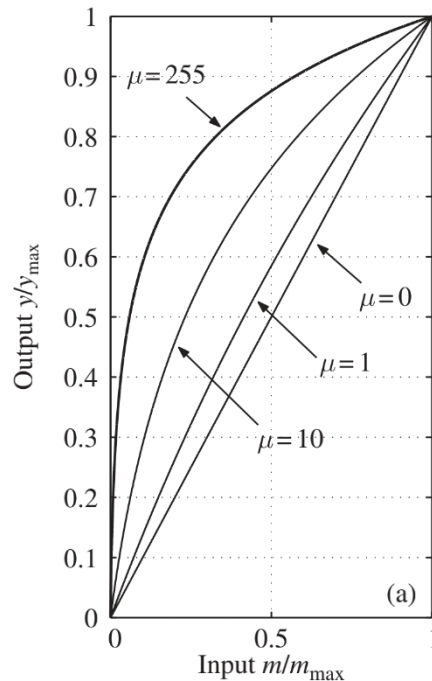
$$q = \begin{cases} q_{max} \frac{A(|m|/m_{max})}{1 + \ln A} \text{sgn}(m), & 0 < \frac{|m|}{m_{max}} \leq \frac{1}{A} \\ q_{max} \frac{1 + \ln [A(|m|/m_{max})]}{1 + \ln A} \text{sgn}(m), & \frac{1}{A} < \frac{|m|}{m_{max}} \leq 1 \end{cases},$$

- The compression characteristic of μ -law is:

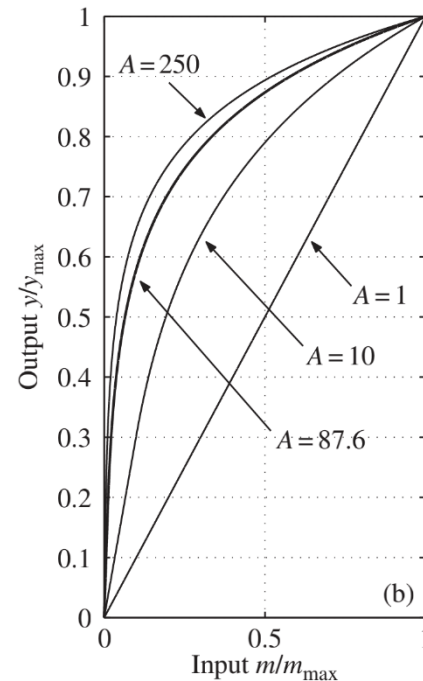
$$q = q_{max} \frac{\ln [1 + \mu(|m|/m_{max})]}{\ln(1 + \mu)} \text{sgn}(m)$$

Robust Quantizer (4)

- **A-law and μ -law**
 - In ITU G.712 Standard:
 - The recommendation value for μ -law compander is $\mu = 255$
 - The recommendation value for A-law compander is $A = 87.6$
 - Compression characteristics of two companders:



μ -law



A-law

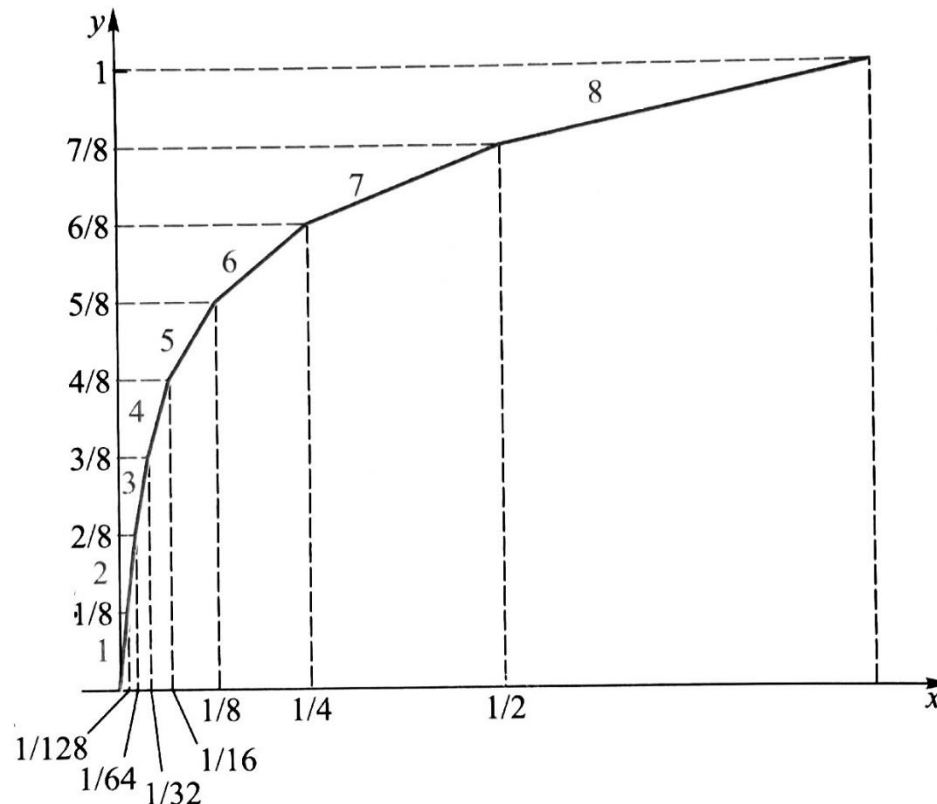
Robust Quantizer (5)

- **Piecewise Approximation**
 - Nonlinear analog circuits are complex to implement
 - It is usual to apply piecewise approximations to original companders
- Two types of widely used piecewise approximations
 - 13-piece A-law: Used in Europe, Africa, China
 - 15-piece μ -law: Used in America, Canada, Japan
- For simplicity, we apply piecewise approximations to the normalized input x and normalized output y :

$$\begin{cases} x = m/m_{max} \\ y = q/q_{max} \end{cases}$$

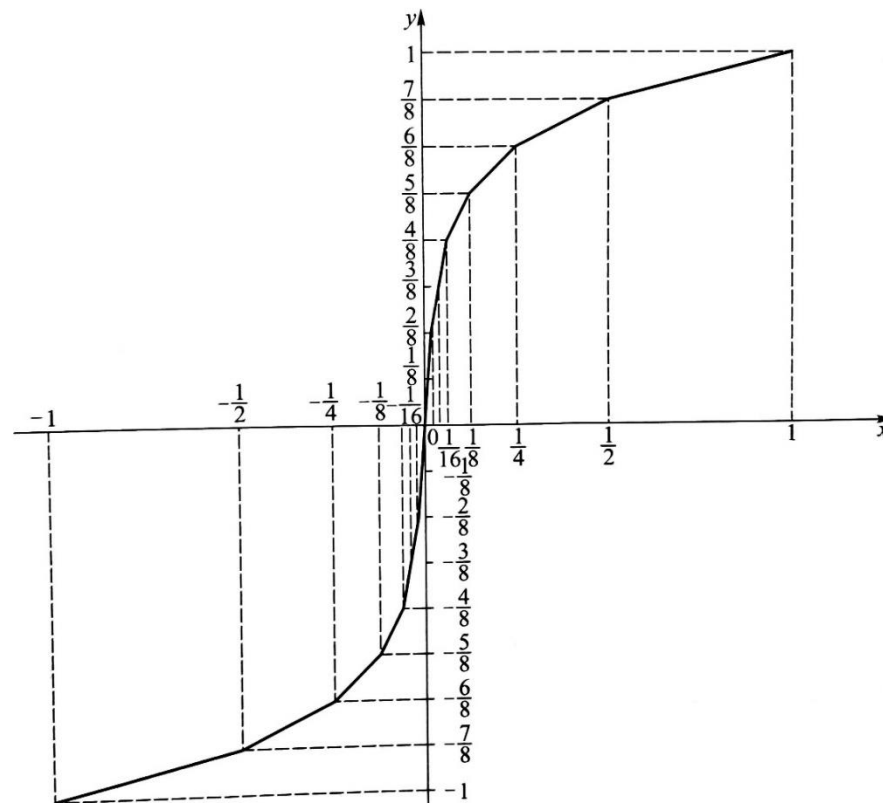
Robust Quantizer (6)

- **Piecewise Approximation for A-law Compander**
 - **13-piece A-law**: the y-axis is uniformly partitioned to 8 segments, while the x-axis is halved for each region:



Robust Quantizer (7)

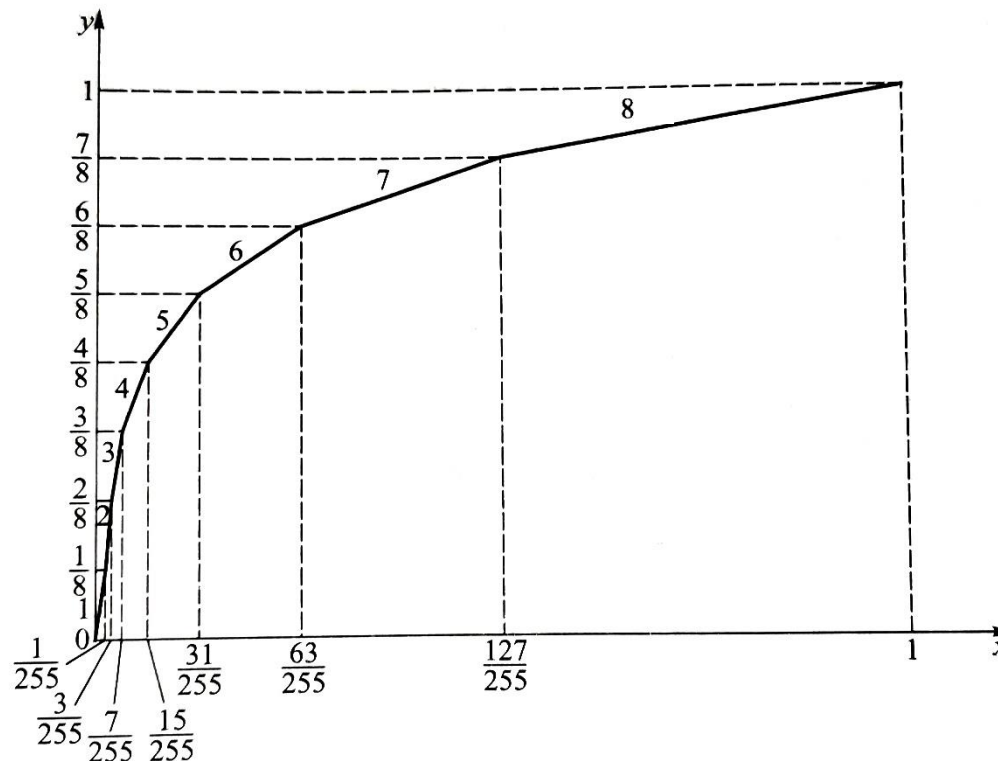
- **Piecewise Approximation for A-law Compressor**
 - **13-piece A-law**: the complete 13-pieces A-law mapping quantizer is an odd function, with four segments in the center merged into one segment



*Robust Quantizer (8)

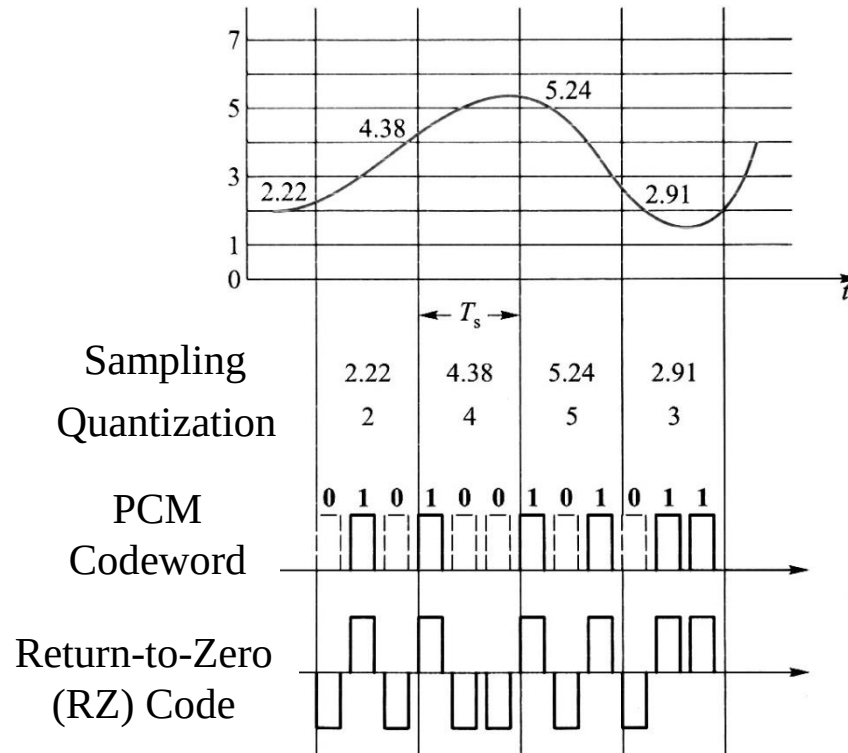
- **Piecewise Approximation for μ -law Compressor**
 - 15-piece μ -law: the y-axis is uniformly partitioned to 8 segments, while the decision levels on x-axis is given by:

$$x = \frac{256^y - 1}{255} = \frac{256^{i/8} - 1}{255} = \frac{2^i - 1}{255}$$



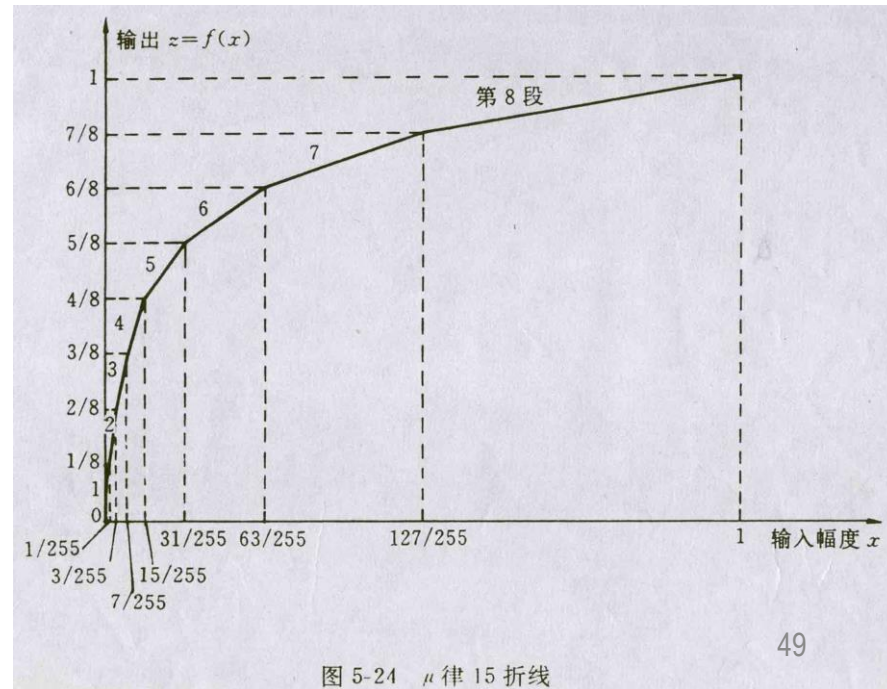
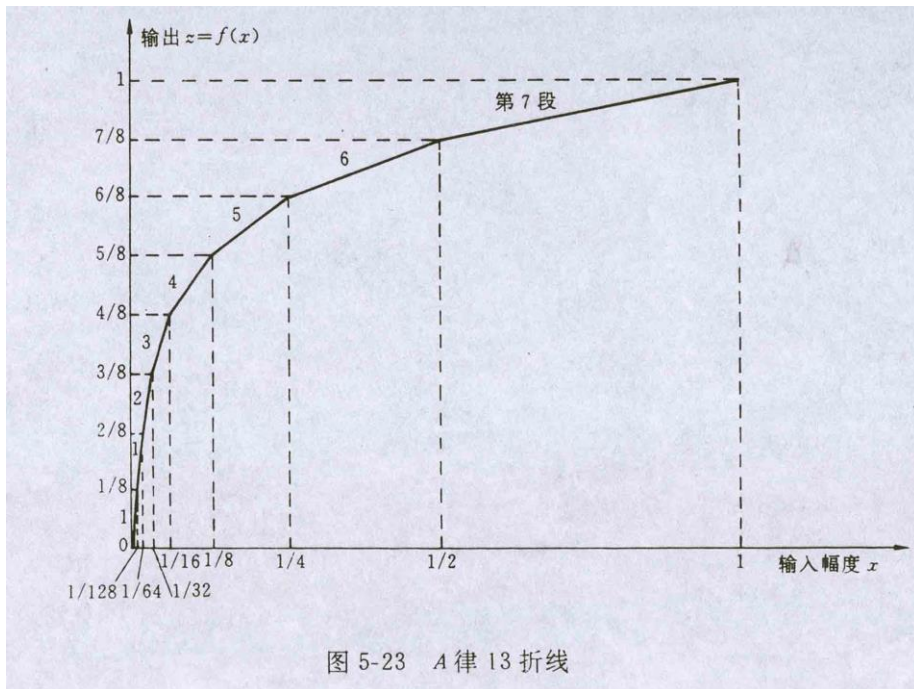
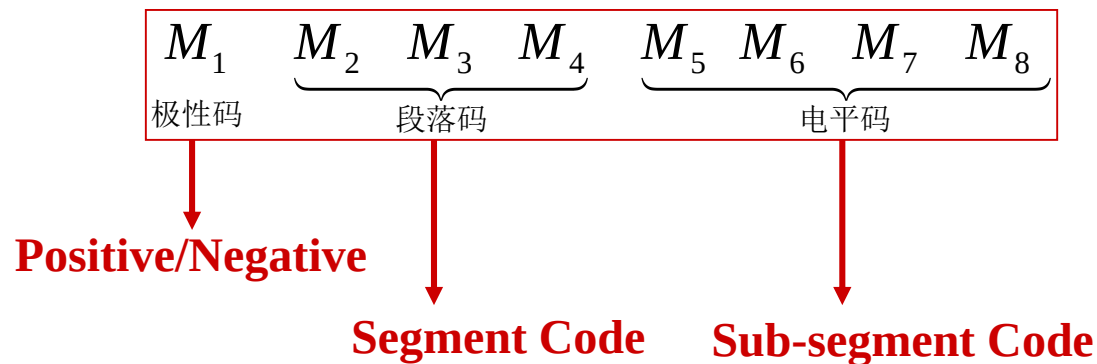
Pulse-Code Modulation (1)

- **Pulse-Code Modulation**
 - Pulse-Code Modulation (PCM) is a whole process of sampling, quantization and encoding to digital codewords
 - From analog signals to binary digits



Pulse-Code Modulation (2)

- Coding Protocol
- 8-bit



Pulse-Code Modulation (3)

$$V_{max} = 4096$$

Segment	Segment code			Start Point	Sub-segment code				Quantization interval
	M ₂	M ₃	M ₄		M ₅	M ₆	M ₇	M ₈	
0	0	0	0	0	16	8	4	2	2
1	0	0	1	32	16	8	4	2	2
2	0	1	0	64	32	16	8	4	4
3	0	1	1	128	64	32	16	8	8
4	1	0	0	256	128	64	32	16	16
5	1	0	1	512	256	128	64	32	32
6	1	1	0	1024	512	256	128	64	64
7	1	1	1	2048	1024	512	256	128	128

PCM code of +1250 ?

Pulse-Code Modulation (4)

- **Code for +1250**

- $1250 > 0$

$$M1 = 1$$

→ Positive

- $1250 > 256$

$$M2 = 1$$

- $1250 > 1024$

$$M3 = 1$$

- $1250 < 2048$

$$M4 = 0$$

} Segment code

- $1024 + 512 > 1250$

$$M5 = 0$$

- $1024 + 256 > 1250$

$$M6 = 0$$

- $1024 + 128 < 1250$

$$M7 = 1$$

- $1024 + 128 + 64 < 1250$

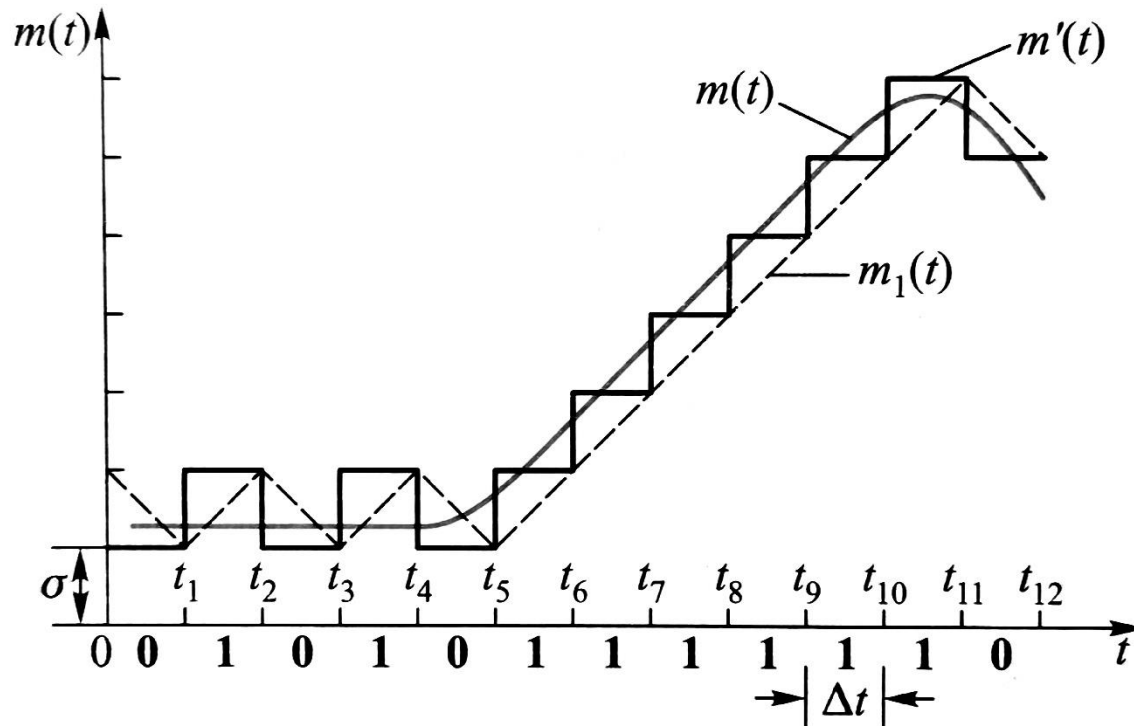
$$M8 = 1$$

} Sub-segment code
32: half of the quantization interval of segment 110

Representation point: $1024 + 128 + 64 + 32 = 1248$

Pulse-Code Modulation (5)

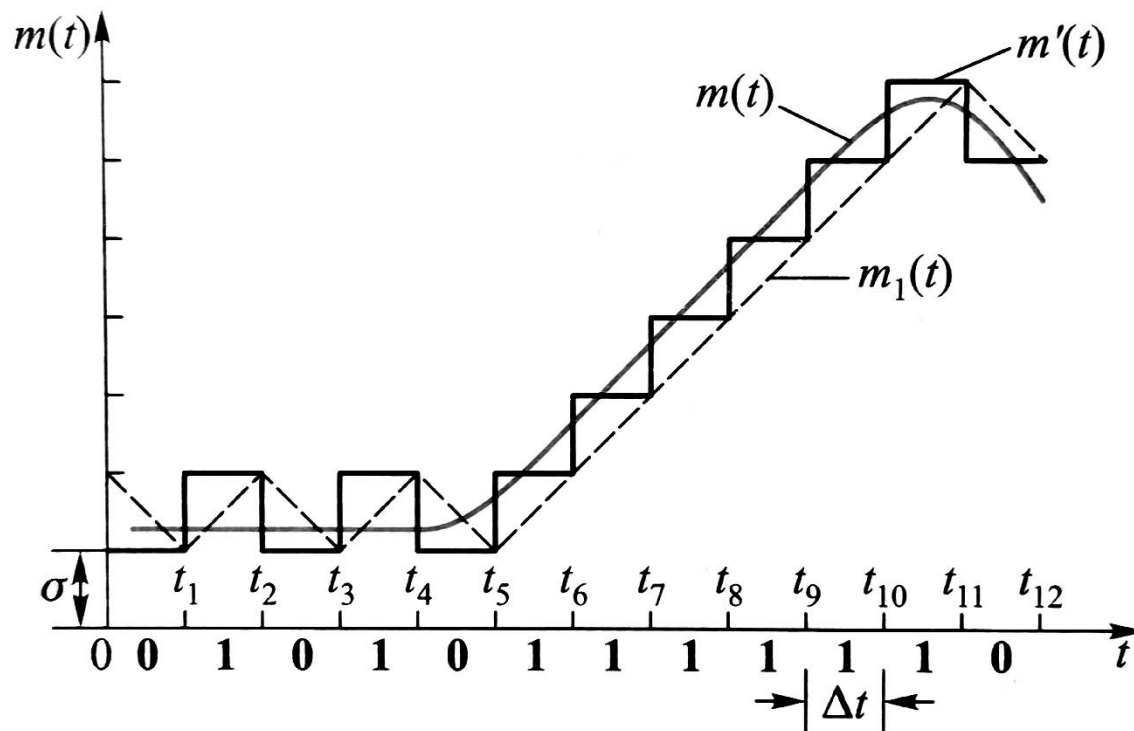
- **Differential Modulation (DM)**
 - When the sampling frequency is high, the amplitude difference between consecutive samples is small
 - Since the range is smaller, it is easier to quantize the amplitude difference rather than the exact values



Pulse-Code Modulation (6)

- **Differential Modulation (DM)**

- Let $m(t)$ be the original continuous signal.
DM uses staircase signal $m'(t)$ to approximate $m(t)$
- Since $m'(t)$ always goes up / down by one stair, $m'(t)$ can be encoded by a binary sequence

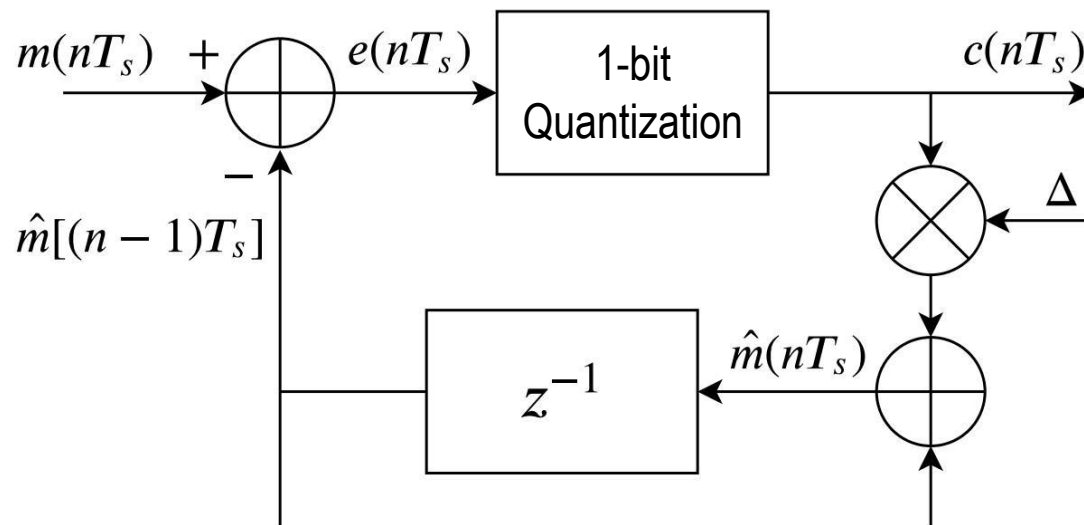


Pulse-Code Modulation (7)

- **Differential Modulation (DM)**

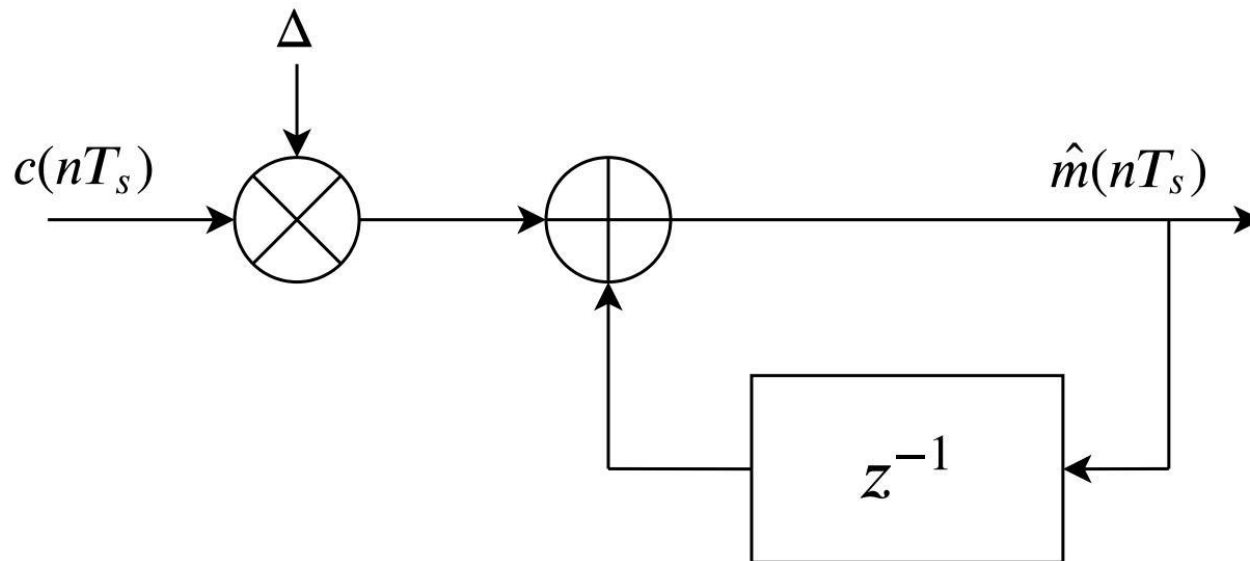
- Diagram of encoder

- $m(nT_s)$: the sampled signal
- $\hat{m}(nT_s)$: the reconstructed signal
- $e(nT_s)$: the differential signal $e(nT_s) = m(nT_s) - \hat{m}[(n-1)T_s]$
- $c(nT_s)$: the binary sequence



Pulse-Code Modulation (8)

- **Differential Modulation (DM)**
 - Diagram of decoder
 - The decoder of receiver and sender is the same
 - Because of the feedback in the encoder, the reconstruction signal $\hat{m}(nT_s)$ tracks the sample signal $m(nT_s)$
 - **DM is robust against bit error and is widely used in warfare communication**



Pulse-Code Modulation (10)

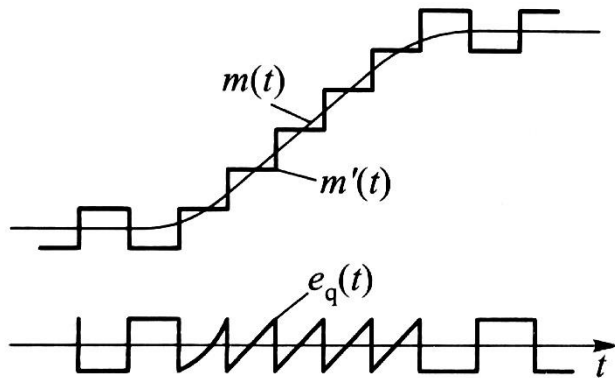
• Differential Modulation (DM)

- DM also brings quantization noises and suffers from overload
- Overload could happen when the signal changes rapidly
- No overload will happen if $m(t)$ satisfies:

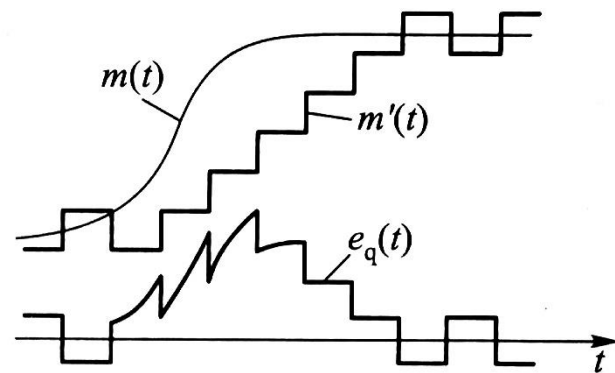
- Δ_t : sampling period

- σ : quantization interval

$$\left| \frac{dm(t)}{dt} \right|_{max} \leq \frac{\sigma}{\Delta_t} \triangleq k_{max}$$



(a) Normal quantization noise



(b) Overload quantization noise

Pulse-Code Modulation (11)

- **Differential Modulation (DM)**

- Example: For the sinusoidal input $m(t) = A\sin(\omega_k t)$,

$$\left| \frac{dm(t)}{dt} \right|_{max} = A\omega_k$$

- Let sample rate $f_s = 1/\Delta_t$, the threshold for overload is

$$A\omega_k \leq \sigma \cdot f_s \Rightarrow A_{max} = \frac{\sigma f_s}{\omega_k} = \frac{\sigma f_s}{2\pi f_k}$$

- For DM system, the maximum load decreases as the frequency increases

- SNR_q deteriorates in the high frequency of voice signal, thus DM cannot be applied directly in voice communication systems

Thank you !

(Q&A)

Reference (Sampling):

[1] H. Nguyen, *A First Course in Digital Communications*, Section 4.1.1

[2] J. Proakis, *Digital Signal Processing*, Section 6.4.1

Reference (Quantization):

[1] H. Nguyen, *A First Course in Digital Communications*, Section 4.3.1-4.3.4